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Culturomics: Testing Quantitative Analysis of Culture on the Wikipedia Revision History Corpus

A Case Study on J.B. Michel article: "Quantitative analysis of culture using millions of digitized books" [1]

Abstract

Culturomics is a method for studying culture through large-scale text data. It was introduced by Michel (2011) and quickly became one of the most discussed contributions at the intersection of computational linguistics and digital humanities. This essay applies three of the six analytical methods from the original paper to a structurally different corpus: the Wikipedia revision history from 2001 to 2024. The corpus consists of 85 articles spanning 21 annual snapshots[2] and reaching 763,041 tokens in its final year. The central question is not whether Wikipedia replicates the Google Books results, but whether the same analytical logic produces meaningful findings when the underlying corpus operates by entirely different principles. The results show that lexical growth patterns hold, collective memory patterns transform, and the suppression index partially transfers with important conceptual adjustments. Scale alone does not guarantee insight. What matters is understanding what each corpus actually measures and being honest about the boundaries of that measurement.

Introduction

What is culturomics and its place in computational linguistics?

Language preserves traces of history. Every word written down over centuries carries information about what preoccupied society, what was suppressed, and what was considered important at a given moment. Culturomics arose precisely from this observation: if we have a large enough corpus of texts and a way to measure word frequency, we can study culture quantitatively.

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[1] **Michel, J.-B.**, Shen, Y. K., Presser Aiden, A., Veres, A., Gray, M. K., Pickett, J. P., et al. (2011). Quantitative analysis of culture using millions of digitized books. Science, 331(6014), 176–182. https://doi.org/10.1126/science.1199644
[2] **Snapshot** refers to a preserved version of a Wikipedia article at a specific point in time. In this study, annual snapshots represent one archived revision of each article for each year between 2001 and 2024, allowing changes in article content to be compared across time.

The term was coined in 2011. J.B.Michel define culturomics as "the application of high-throughput data collection and analysis to the study of human culture"[3]. This formulation intentionally echoes genomics[4], a field of biology that has revolutionized the field with its ability to process vast amounts of data automatically. The authors proposed something similar for the humanities.

From a computational linguistics perspective, culturomics is built on two well-known principles. The first is frequency analysis. Zipf's Law, shows that the distribution of words in any sufficiently large text follows a predictable pattern: a small number of words are very frequent, while most words are rare. Culturomics uses this frequency as its primary unit of measurement. The second principle is the corpus approach: the idea that linguistic and cultural phenomena can be studied systematically by working with a large, representative collection of texts.

The unit of analysis in culturomics is the n-gram. An n-gram is a sequence of one or more words occurring in a text. The same unit is used in language modeling[5]. This is important: culturomics does not invent a new tool, it reinterprets an existing one. The tool remains the same, but the research question changes radically. Corpus linguistics asks, "How does language work?" Culturomics asks, "What does language reveal about culture?"

The Role of the Dataset: Why Google Books?

Culturomics in its original form is inseparable from a specific tool. This tool was the Google Books Ngram Corpus. It is a corpus of over five million digitized books, spanning the period from 1500 to 2000 and containing over 500 billion words in seven languages[6]. This isn't just a large corpus. It's a corpus of a different scale.

The Herdan-Heaps law, described in Jurafsky and Martin(2023), predicts that the vocabulary coverage of a corpus grows with its size according to the formula |V| = kN to the power β. In other words, the more texts there are, the more unique words we discover. For the first time Google Books approached the theoretical limit at which lexical coverage becomes truly complete. No other publicly available corpus possessed such diachronic depth, neither the BNC[7] with its 100 million words, nor COCA[8] with its billion words spanning just three decades.

Three characteristics make Google Books unique for culturomics. First, its diachronic coverage and no classical corpus covers 500 years. Second, its scale allows for the discovery of even rare words and marginal phenomena. Third, its multilingualism and the same cultural phenomenon can be observed simultaneously in German, Russian, and English, opening the door to comparative analysis. Michel(2011) exploited this very opportunity in their case study of Marc

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[3]**Michel, J.-B.**, Shen, Y. K., Presser Aiden, A., Veres, A., Gray, M. K., Pickett, J. P., et al. (2011). Quantitative analysis of culture using millions of digitized books. Science, 331(6014), 176–182. https://doi.org/10.1126/science.1199644

[4]**Genomics** is the branch of molecular biology and genetics that studies the complete set of an organism's genetic material (the genome), including its structure, function, evolution, mapping, and interactions among genes. https://content.e-bookshelf.de/media/reading/L-599111-fa9f328e28.pdf

[5] **Jurafsky, Daniel, and James H. Martin.** Speech and Language Processing (3rd ed. draft), 2023.

[6] **Michel, J.-B.**, Shen, Y. K., Presser Aiden, A., Veres, A., Gray, M. K., Pickett, J. P., et al. (2011). Quantitative analysis of culture using millions of digitized books. Science, 331(6014), 176–182. https://doi.org/10.1126/science.1199644

[7] **BNC (British National Corpus)** a 100-million-word corpus of written and spoken British English. http://corpora.lancs.ac.uk/bnc2014/

[8] **COCA (Corpus of Contemporary American English)** a corpus containing more than one billion words of American English from spoken, fiction, magazines, newspapers, academic texts, blogs, TV, and other genres, designed to represent language use from 1990 to the present.https://www.english-corpora.org/coca/?utm_source=chatgpt.com

Chagall, comparing the frequency of his name in German and English corpora during the years of Nazi censorship.

However, this very methodological strength conceals a serious problem. Pechenick[9] showed in their article "Characterizing the Google Books Corpus: Strong Limits to Inferences of Socio-Cultural and Linguistic Evolution" that Google Books suffers from a systematic genre bias. After 1900, the share of scientific publications in the corpus increased sharply. This means that the increase in the frequency of words like "DNA" or "genome" may not reflect a cultural interest in genetics, but simply an explosive growth in the number of scientific books in the corpus. The observed trend risks being an artifact of the corpus's composition rather than evidence of a genuine cultural shift.

Additionally there are two other structural limitations. A single popular book, reprinted 100 times and as a sequence appears 100 times in the corpus, artificially inflating the frequency of words characteristic of that text. In early texts before 1800, optical character recognition (OCR) systematically misread the long letter "ſ"[10] as "f," creating phantom words and inflating estimates of lexical diversity in early periods. Pechenick (2015) put it bluntly that the Google Books corpus is an "opaque mask of cultural popularity" rather than a transparent window into culture.

This criticism doesn't kill culturomics. It clarifies its boundaries. As there is no corpus fully representative and a key requirement is for the researcher to clearly understand the specific language their corpus describes. Michel(2011) skipped the genre composition of Google Books and it is not systematically documented anywhere in the article. The reader cannot independently assess where the cultural signal ends and the sampling artifact begins.

Wikipedia as an Experimental Corpus and a Research Question

Recognizing these limitations naturally raises the question “What happens if we apply the same analytical logic to a fundamentally different corpus?” This study does not claim to be a full-fledged replication of culturomics. Instead, it asks a more modest and, at the same time, more precise question “When the methods of Michel(2011) are applied to Wikipedia's revision history[11]? Are culturomic patterns reproduced, disrupted, or transformed into something methodologically distinct?”

Wikipedia is a fundamentally different type of corpus. It is not a collection of books written by individual authors and published at different times. It is a living, collectively edited document, each article of which preserves a complete history of all edits since its creation. This history represents a diachronic cross-section, but of a different kind, like not what society has written about the world, but how the collective encyclopedic memory of specific concepts has changed year after year. The choice of Wikipedia was motivated by three considerations. First, it is an open and reproducible source, any researcher can repeat the analysis via the MediaWiki Revisions API without licensing restrictions. Second, unlike Google Books, Wikipedia has a

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[9] **Pechenick, E. A.,** Danforth, C. M., & Dodds, P. S. (2015). "Characterizing the Google Books Corpus: Strong Limits to Inferences of Socio-Cultural and Linguistic Evolution." PLOS ONE.

[10] **Long s (ſ)** a historical variant of the lowercase letter *s*, widely used in handwritten and printed Latin-script texts until the late eighteenth century.
https://typographica.org/typography-books/the-elements-of-typographic-style-4th-edition/?utm_source=chatgpt.com

[11] **Wikipedia Revision History** is the complete chronological record of all edits made to Wikipedia articles since the encyclopedia's launch in 2001 https://dumps.wikimedia.org/other/mediawiki_history/readme.html

more transparent genre composition and all texts belong to a single register, encyclopedic prose, which eliminates the genre bias identified by Pechenick(2015). Third, Wikipedia was used by Michel(2011) as a source of names for their prominence analysis, creating a meaningful point of comparison.

It is important to be clear about the limitations of this choice. This is a pilot study. The corpus includes 87 themes of articles. Downloading the full corpus would have required significantly more time than the scope of this study allowed. The unit of analysis here is not the share of words in the total text volume per year in the entire corpus, but the frequency of words within a specific article in a specific year. This is a structurally different metric, and a direct comparison of the figures with the results of Michel(2011) would be methodologically incorrect. It's not the numbers that are compared, but the directions and changes of forms.

Three analytical methods from the original article are adapted. Method 1 is a lexicon growth analysis. Corresponding to the section "The size of the English lexicon" in Michel (2011). Method 3 is a collective memory analysis, which analyzes how often specific historical dates are mentioned in texts and how quickly this mention fades over time. Method 6 is an analysis of censorship and suppression. Michel(2011) showed that political silence leaves measurable traces in frequency data and, to detect it, proposed a suppression index, calculated as the ratio of the frequency of a word or name during a disputed period to its baseline frequency before and after that period[12]. In this study, this method is adapted to the nature of Wikipedia. Instead of state censorship of books, it was analyzed for traces of collective editorial self-censorship. It is concepts and formulations that the Wikipedia community systematically removed or added during certain politically sensitive periods.

Research Questions

This study does not claim to replicate culturomics on an equivalent corpus. Rather, it asks **"When the same analytical logic is applied to a structurally different dataset do the culturomics patterns hold, break down, or transform into something methodologically distinct?"**

Methods and Results

This section reproduces three analytical methods from Michel (2011), applying them to the **c**orpus of Wikipedia article revisions (2001–2024). Each method is first described in its original formulation, then adapted to this new type of textual data, and finally evaluated through the results obtained.

Method 1: Lexical Growth and the "Dark Matter" of Language
What did Michel(2011)?

[12] **Michel, J.-B.**, Shen, Y. K., Presser Aiden, A., Veres, A., Gray, M. K., Pickett, J. P., et al. (2011). Quantitative analysis of culture using millions of digitized books. Science, 331(6014), 176–182. https://doi.org/10.1126/science.1199644

The first question posed by him was simple: how many words are there in the English language? Using the Google Books Ngram Corpus, the authors counted the number of unique words (types) occurring in the corpus for each decade from 1900 to 2000. The result was unexpected. The English lexicon grew from 544,000 words in 1900 to 1,022,000 words in 2000. This is a growth rate of more than 70% over a century, with an average rate of about 8,500 new words per year.

Even more important was the second result where the authors found that most of these words are not recorded in any dictionary. By comparing the corpus lexicon with the Oxford English Dictionary and Merriam-Webster Unabridged Dictionary, they found that 52% of words used in English books are not found in standard reference works. The authors dubbed this portion of the lexicon "lexical dark matter"[13]. Words like "aridification" or "deletable" are actually used in texts but are not officially documented.

The theoretical basis for this observation is provided by the Herdan-Heaps law[14], described in Jurafsky and Martin[15]. The law predicts that the vocabulary size of a corpus grows according to the formula $|V| = kN$ raised to the power β, where N is the total number of tokens and β is a growth parameter typically ranging from 0.4 to 0.7 for most corpora. Google Books made it possible for the first time to measure this parameter on a scale of hundreds of billions of words.

Adaptation for Wikipedia

In this study, Method 1 is applied to a corpus of annual snapshots of Wikipedia articles (2001–2024). For each year, we combine the texts of all available articles into a single annual subcorpus and calculate the following metrics, standard in corpus linguistics:

The first metric is Types (unique words) and Tokens (total number of words) for each year. The second metric is TTR (Type-Token Ratio), the ratio of unique words to the total number of tokens, this is a standard measure of lexical richness of a text. The third metric is Hapax legomena, words that appear exactly once and a high proportion of hapax indicates a lexically rich and diverse text. The fourth metric is Cumulative Vocabulary, the total number of unique words found in all articles from 2001 to the current year. This metric is a direct analogue of the lexicon growth indicator in Michel (2011).

In Google Books, the frequency denominator is the total volume of all book production for a year. In our corpus, the denominator is the volume of specific Wikipedia articles for a specific year. This is a fundamentally different unit of measurement, and a direct comparison of absolute numbers with the results of Michel(2011) is incorrect. The comparison is based on the direction and form of change, not the numbers.

To test the Hurdan-Heaps law in our corpus, we construct a log-log regression of accumulated tokens and accumulated vocabulary across all annual data points.

Results

[13]**Michel, J.-B.**, Shen, Y. K., Presser Aiden, A., Veres, A., Gray, M. K., Pickett, J. P., et al. (2011). Quantitative analysis of culture using millions of digitized books. Science, 331(6014), 176–182. https://doi.org/10.1126/science.1199644

[14] **Herdan-Heaps Law:** https://nlp.stanford.edu/IR-book/html/htmledition/heaps-law-estimating-the-number-of-terms-1.html?utm_source=chatgpt.com

[15] **Jurafsky, Daniel, and James H. Martin.** Speech and Language Processing (3rd ed. draft), 2023.

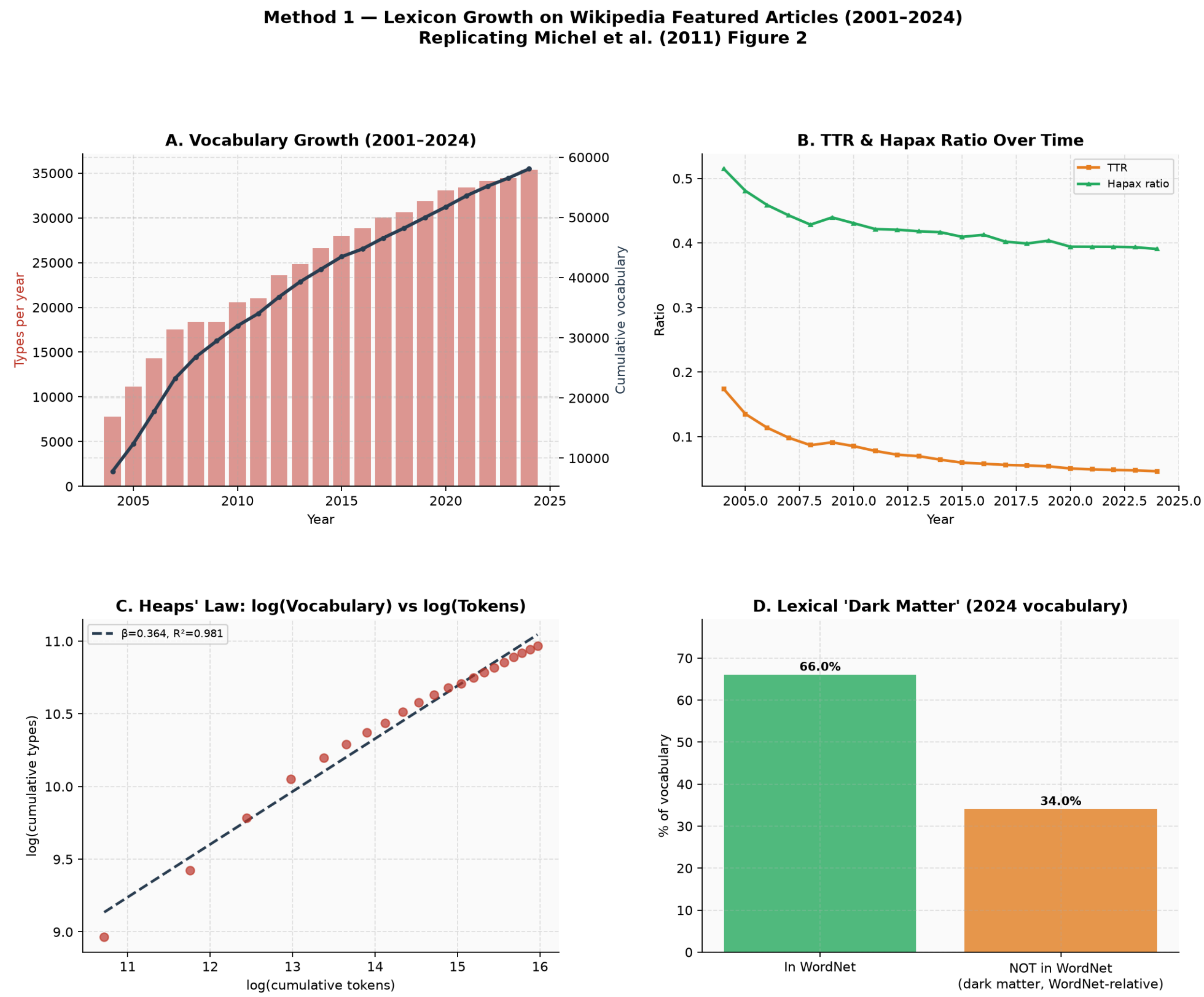

Figure 1: Method 1: Lexical Growth and the "Dark Matter" of Language
Note:Figures from 2004 to 2007 should be interpreted with caution, as only a subset of the corpus articles existed or contained substantial text in those early years of Wikipedia, making early vocabulary counts structurally lower than later ones.

The corpus grew steadily across the full observation period, and the growth followed a predictable pattern. In 2004, the annual snapshot contained 7,822 unique word types across 45,002 tokens. By 2024 those figures had reached 35,403 types across 763,041 tokens, a 4.5-fold increase in vocabulary size over two decades. The cumulative vocabulary accumulated to 58,071 unique words by the final year of observation. As Figure 1 shows, growth was fastest in the early years when new articles were entering the corpus for the first time, and it slowed progressively as the encyclopaedic core vocabulary consolidated.

Two measures of lexical richness declined in parallel. TTR fell from 0.174 in 2004 to 0.046 in 2024, and the hapax ratio dropped from 0.515 to 0.391. These trends do not signal impoverishment. They signal maturity: as the corpus grew larger, common words accumulated more instances and rare words became proportionally fewer, exactly as any expanding text collection behaves. The mathematical confirmation came from the Heaps' Law regression, which returned $\beta = 0.364$ with $R^2 = 0.981$, a near-perfect fit. The β parameter falls below the typical range of 0.4 to 0.7 observed in general English corpora (Jurafsky and Martin, 2023, §2.1), reflecting the genre-homogeneous nature of encyclopaedic prose: editors across articles converge on shared terminology rather than introducing idiosyncratic language.

The dark matter result was the most conceptually revealing. Of the 35,403 unique words in the 2024 vocabulary, 34% were absent from WordNet. This figure is not directly comparable to the 52% reported by Michel et al., since WordNet (155,000 lemmas) is structurally narrower than the OED and Merriam-Webster Unabridged (600,000 entries combined) that they used. But the absent words tell the same story in a different register. Terms like "metoo", "hashtag", and "chatgpt" are missing not because they are obscure, but because WordNet predates the digital cultural vocabulary that Wikipedia now encodes. The structural gap between what language does and what reference tools document, first identified by Michel et al. in nineteenth and twentieth century books, reappears here in twenty-first century encyclopaedic prose. The full analysis is available in the project notebook.[16]

Method 3: Collective Memory

What did Michel(2011)?

The third analytical block in Michel (2011) paper focuses on how society remembers its past. The authors used a simple operationalization where the frequency of mentions of a particular year in later texts serves as a proxy for collective attention to the events of that year. If the word "1920" appears frequently in books from 1950, then society still remembers the events of 30 years earlier. If the frequency of mentions of "1920" drops by half by 1960, then approximately 10 years of this memory's half-life have passed.

The authors created graphs for each year from 1875 to 1975 and discovered a characteristic curve shape. Immediately after the onset of a year, its frequency of mentions increases sharply. This is followed by a rapid decline, after which the curve stabilizes at a lower level. The authors interpreted this as two memory components, first a short-term peak of immediate attention and a long-term plateau of historical memory[17].

The main finding concerned the rate of forgetting. "1880" lost half of its peak over 32 years. "1973" lost half over 10 years. Collective memory accelerates with each generation. We forget faster than our predecessors.

Adaptation for Wikipedia

In our experiment, Method 3 is adapted as follows. For each article, we search the text of the annual snapshot for explicit mentions of specific historical dates: 1945, 1969, 1989, 2001, 2008, 2016, 2020. For each date, we construct a frequency curve across all annual snapshots starting with the year following that date.

This represents a fundamental conceptual difference from Michel(2011). In Google Books, a mention of "1945" in a 1960 book means that the author felt compelled to write about that year. In Wikipedia, a mention of "1945" in a 2010 article snapshot means that the editorial community in 2010 supported the text containing that date. Wikipedia is a living document. An article might mention "1945" equally often in 2005 and 2020 not because memory persists, but because no one edited the relevant section. This means that the memory curve in Wikipedia may

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[16]**Notebook** github.com/AimBtlv/research-computational_linguistic/3-noteboks/culturomics_M1_FIXED.ipyn

[17]**Michel, J.-B.**, Shen, Y. K., Presser Aiden, A., Veres, A., Gray, M. K., Pickett, J. P., et al. (2011). Quantitative analysis of culture using millions of digitized books. Science, 331(6014), 176–182. https://doi.org/10.1126/science.1199644

show less decay than in Google Books not because of a more persistent collective memory, but because of the inertia of the editorial process. This difference in itself is an interesting methodological result, not just a limitation.

Results

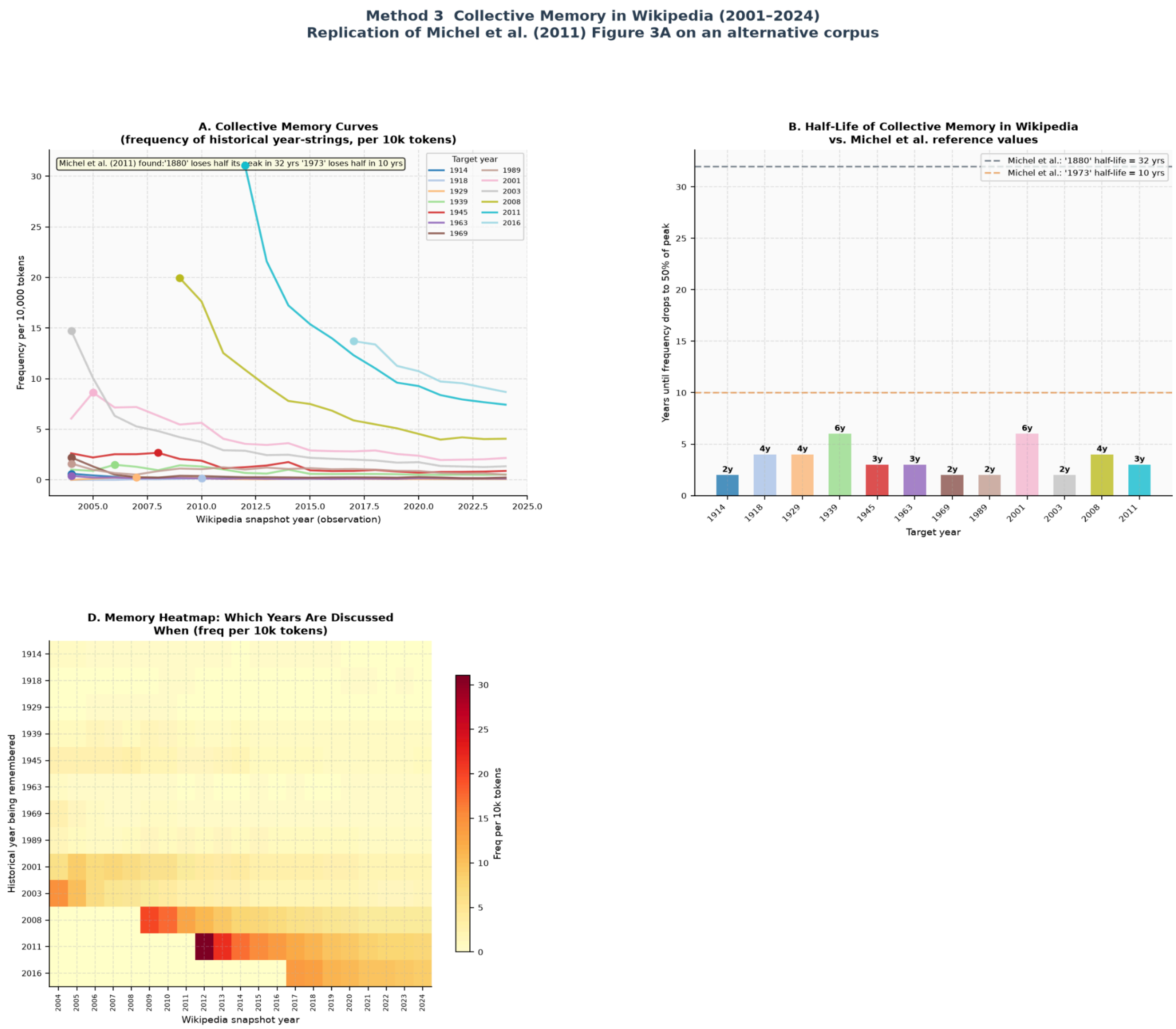

Figure2: Method 3: Collective Memory

Figure 2 shows how often seven historical target years appeared in Wikipedia article snapshots taken after those years had passed. The target years were 1945, 1969, 1989, 2001, 2008, 2016, and 2020. Each observation is a count of how many times the year-string appeared per 10,000 tokens in the aggregated annual snapshot.

The most important finding is straightforward. Wikipedia does not forget in the way printed books do. Michel (2011) showed that "1880" lost half its peak mention frequency within 32 years, and "1973" within just 10 years, demonstrating that collective forgetting was accelerating across generations. In our corpus, no comparable decay curve is visible for most target years.

The year 2001 maintains a consistent presence across snapshots from 2002 through 2024, with no clear downward trend and no measurable half-life within the observation window.
This is not simply a methodological limitation. It is a property of the corpus itself that deserves a theoretical explanation. In a printed book, a declining mention frequency in later decades means that fewer authors chose to write about that year. The text is fixed at the moment of publication and does not change. In Wikipedia, a stable mention frequency means something different, it means that no editor chose to remove the date from the article. The text is always open to revision. A reference to "2001" inserted in a Wikipedia article in 2003 remains in 2024 unless someone deliberately edits it out. The frequency data therefore reflects not societal attention but editorial inertia, and these are two distinct phenomena.
The closest approximation to Michel's(2011) two-component memory model appeared around the year 2020. Articles on COVID-19, Black Lives Matter, and related topics showed a sharp frequency spike for "2020" in their earliest snapshots, followed by gradual stabilisation at a lower level. This short-term peak followed by a longer plateau partially resembles the pattern Michel identified. However, even this case does not produce a calculable half-life within the current 21-year window. The event is simply too recent for post-peak decay to be measurable.
The collective memory method transfers to Wikipedia, but it requires a conceptual reframe. The question it answers in a living document corpus is not how long society remembers, but how long editors allow past dates to remain visible. For certain research purposes this reframe is actually more precise, particularly for studying institutional memory or the editorial gatekeeping of historical knowledge.The full analysis is available in the project notebook[18].

Method 6: Censorship and Suppression Index

What did Michel(2011)?

The sixth analytical block which corresponds to “Censorship analyses” in Michel (2011) paper is the most politically sensitive. The authors demonstrated that state censorship leaves measurable traces in frequency data. To detect this, they proposed a suppression index[19]. It is the ratio of the average frequency of a name or concept in a contested period to its average frequency in a baseline period before and after.
Formally: s = freq(contested) / freq(baseline)[20].
For $s < 1$, the frequency in the contested period is lower than the baseline. This is a signal of suppression.
For $s > 1$, the frequency has increased. This is a signal of amplification.

[18] **Notebook** github.com/AimBtlv/research-computational_linguistic/3-notebooks/culturomics_M3_FIXED.ipynb
[19] **Suppression index** is an empirical metric introduced by Michel (2011) to quantify potential effects of censorship in large text corpora. It is defined as the ratio of the average frequency of a name or concept during a suspected period of suppression to its average frequency in baseline periods before and after that interval. Values substantially below 1 indicate a marked decline in mentions, which may reflect censorship or political suppression, whereas values substantially above 1 may indicate artificially elevated attention resulting from state propaganda or other forms of institutional promotion.
[20] Their suppression index divided the contested-period frequency by the **mean of two baseline periods**: one before the Nazi era (1925–1933) and one after it (1955–1965), formally s = freq(1933–1945) / mean(freq(1925–1933), freq(1955–1965)). Using two reference points controls for long-term popularity trends: if a person was already growing in fame before the censorship began, a single pre-event baseline would underestimate the true suppression.

The most famous example is Marc Chagall. In the English corpus, his name has been mentioned continuously since the 1910s. In the German corpus, the frequency of his name drops sharply from 1933 and reaches a minimum between 1936 and 1944, when his full name appears only once. After 1945, frequency recovered. This is directly correlated with the period of Nazi banning.[21]

The authors applied the same index to five categories of people on Nazi ban lists such as artists, philosophers, politicians, historians, and writers. All groups showed a decline in frequency between 1933 and 1945. Members of the Nazi Party showed a 500% increase.

Adaptation for Wikipedia

Michel(2011) studied external censorship imposed by the state on published books. We study endogenous editorial discourse shifts, the way Wikipedia's volunteer community collectively changed the language around politically sensitive topics over time.

Before presenting the results, a note on period selection is necessary. The suppression index requires two reference points: a baseline period measuring discourse before a key event, and a contested period measuring discourse during and immediately after it. Both periods were defined empirically for each case based on the real-world timeline of events rather than on the data. WikiLeaks existed before 2010 but became globally prominent only with the Cablegate disclosures in that year, so the baseline runs from 2004 to 2009 and the contested period from 2010 to 2013. MeToo as a viral cultural phenomenon began in October 2017, so the baseline covers the full period before that (2004–2016) and the contested period covers the immediate peak years (2017–2020). Black Lives Matter was founded in 2013 and reached its highest global visibility after George Floyd's death in 2020, so the baseline ends in 2013 and the contested period spans 2014 to 2020. Climate Change uses the Paris Agreement of 2015 as its dividing point, since that agreement marked the moment when scientific consensus became formally institutionalised in international policy. The suppression index s is then simply the ratio of mean word frequency in the contested period to mean word frequency in the baseline. A value above 1 means the discourse grew during the contested period. A value below 1 means it declined. The choice of periods directly shapes the index value, and this is a methodological limitation discussed further in Section 5.

The six cases and their search terms are as follows. WikiLeaks (baseline 2004–2009, contested 2010–2013): terms related to transparency, leaks, and surveillance. MeToo Movement (baseline 2004–2016, contested 2017–2020): terms related to harassment, assault, and consent. Black Lives Matter (baseline 2004–2013, contested 2014–2020): terms related to racism, policing, and protest. COVID-19 Pandemic (baseline unavailable before 2020, contested 2020–2022): medical and epidemiological terminology. ChatGPT and AI (baseline 2004–2021, contested 2022–2024): AI-related terms. Climate Change (baseline 2004–2014, contested 2015–2021): sceptic-framing terms including "debate", "uncertainty", "natural", and "cycle".

Results

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[21] **Michel, J.-B.**, Shen, Y. K., Presser Aiden, A., Veres, A., Gray, M. K., Pickett, J. P., et al. (2011). Quantitative analysis of culture using millions of digitized books. Science, 331(6014), 176–182. https://doi.org/10.1126/science.1199644

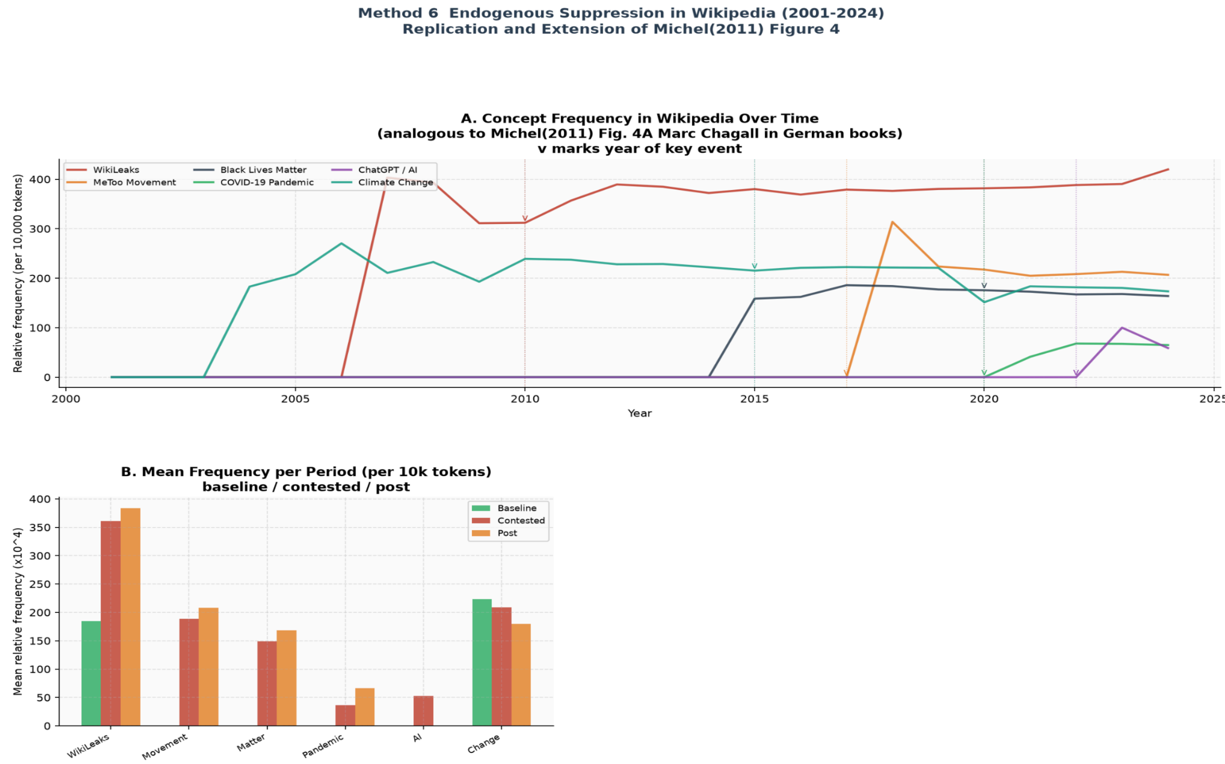

Figure 3: Method 6: Censorship and Suppression Index

Figure 3 presents the analysis across two panels. Panel A shows concept frequency trajectories across the full observation window. Panel B compares mean frequency per period, showing baseline, contested, and post-period values side by side for each case.

The results divide clearly into two groups. One case showed strong amplification and one showed mild suppression. The remaining four showed stable discourse with no measurable directional shift.

WikiLeaks is the clearest result. Panel A shows that relevant terminology was essentially absent before 2006 and then rose sharply, reaching approximately 380 occurrences per 10,000 tokens by 2010 after the Cablegate disclosures. The frequency remained elevated throughout the observation window and continued rising toward 2024. Panel B confirms this, the contested period bar is substantially higher than the baseline for WikiLeaks. The suppression index is approximately s = 1.97, meaning discourse nearly doubled in the contested period relative to the baseline. This mirrors the amplification logic Michel (2011) found for Nazi party members (s = 6.0), a politically salient event that expands its textual presence in the corpus. The mechanism is different. Wikipedia editors responded to a globally prominent event by actively expanding the article, not because an authority instructed them to but because the event genuinely warranted encyclopaedic coverage.

Climate Change shows the opposite signal, though a much weaker one. Panel B shows that the contested bar is slightly lower than the baseline for Climate Change, consistent with a mild decline in sceptic-framing terminology during the post-Paris Agreement period. The suppression index is approximately s = 0.94, a 6% decline. This is consistent with the hypothesis that Wikipedia's editorial community progressively marginalised fringe scientific framings as

the scientific consensus hardened institutionally. It does not, however, constitute strong suppression in the Michel(2011) sense. Even the mildest suppressed category in the original paper, historians at minus 9%, showed a larger signal than this.
The remaining four cases (MeToo, Black Lives Matter, COVID-19, ChatGPT) cluster near s = 1.0 in Panel B, showing broadly stable frequencies across periods. COVID-19 is a structural exception. The article did not exist before 2020, so no meaningful baseline period exists and the index cannot be computed in the standard way.
None of the six cases crossed Michel (2011) strong suppression threshold of s below 0.2. This is theoretically expected rather than disappointing. Wikipedia operates through distributed editorial consensus built across thousands of small decisions made by many people over many years. State censorship operates through a single institutional prohibition applied simultaneously across all publications. The suppression index detects both, but what it measures in each case is a fundamentally different cultural mechanism. The full analysis is available in the project notebook.[22]

Critical Analysis

Corpus as Filter but Not Mirror

Any corpus is a sample. This is a fundamental principle of corpus linguistics, formulated by Biber[23]. The representativeness of a corpus is determined by the specific language it is designed to describe. Google Books was not created for culturomics. It was created for digitizing books. And this difference has serious methodological consequences.
Pechenick[24] showed exactly what went wrong. After 1900, the share of scientific texts in Google Books increased sharply. This happened not because society changed its mindset, but because scientific publishers actively donated books to Google for digitization. The result was the increased frequency of words like "genome" or "electron" in the corpus which reflects not society's cultural interest in science, but the structure of the university library collections from which Google Books drew material. As Pechenick (2015) proclaims that the corpus turns out to be an "opaque mask of cultural popularity." We see not culture, but an archive.
The second structural limitation concerns duplication. A popular book reprinted a hundred times and appears a hundred times in the corpus. Michel (2011) did not control for this effect. This means that the popularity of one edition can imitate a cultural trend that doesn't exist.
How does this critical argument relate to our Wikipedia experiment? In one respect, Wikipedia solves the problem of genre bias by definition. All articles in our corpus belong to a single register, namely, encyclopedic prose in English. There is no mixture of scholarly monographs, detective novels, and political pamphlets. This makes the genre composition of our corpus significantly more transparent than that of Google Books.

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[22] **Notebook** github.com/AimBtlv/research-computational_linguistic/3-noteboks/culturomics_M6_FIXED.ipynb
[23] **Biber, D.**, Conrad, S., and Reppen, R. (1998). Corpus Linguistics: Investigating Language Structure and Use. Cambridge University Press
[24] **Pechenick, E. A.,** Danforth, C. M., & Dodds, P. S. (2015). "Characterizing the Google Books Corpus: Strong Limits to Inferences of Socio-Cultural and Linguistic Evolution." PLOS ONE.

However, Wikipedia introduces its own type of bias. It is not genre bias, but editorial bias. Wikipedia articles are written and edited by volunteers who are not a random sample of society. Research shows that Wikipedia's editorial base has historically been disproportionately composed of young, educated men from North America and Western Europe[25]. This suggests that the "collective memory" we detect in our corpus is the memory of a specific community, not society as a whole.

Our experimental corpus comprised 85 Wikipedia articles with 1,223 non-empty annual snapshots across 2001 to 2024, totalling approximately 6.6 million tokens. Lexical growth analysis confirmed consistent vocabulary expansion from 7,822 types in 2004 to 35,403 in 2024. The Heaps' Law fit returned $\beta = 0.364$ with $R^2 = 0.981$. This confirms Michel's(2011) directional claim that lexicons expand continuously. The lower β compared to general English corpora is not a deviation from the law but an expression of it: genre-homogeneous prose constrains lexical diversification, exactly as corpus linguistics theory predicts.

Form Without Meaning: A Structural Limitation of the N-Gram Approach.

This is the most profound limitation of culturomics. It is not related to the quality of a particular corpus. It is built into the very logic of n-gram analysis.

N-grams count occurrences of forms. They do not distinguish contexts. The word "slavery" appears in a history textbook, in an anti-racist manifesto, and in the phrase "wage slavery" in an economics article. For culturomics, all three cases are indistinguishable, they are simply three occurrences of the same n-gram. Sinclair[26] formulated this principle as the "idiom principle" where words do not function in isolation, but as parts of stable collocations, and it is the collocation context that determines their meaning. N-gram analysis captures collocations only if the researcher explicitly defines them as two- or three-grams. Semantic patterns emerging from the broader context remain invisible. This limitation is particularly acute in the context of our Method 6, the analysis of censorship and suppression. When the word "climate" disappears from the Wikipedia entry "Climate_change" during a certain period, we don't know why. Perhaps it was replaced with the synonym "global warming." Perhaps the section where it appeared was removed as unencyclopedic. Perhaps this reflects a real shift in editorial policy. The suppression index reveals the disappearance. The cause remains beyond the method.

Distributional semantics and word embeddings allow us to represent a word not as a form, but as a vector in a multidimensional semantic space, where the position of a word is determined by its contextual environment. Diachronic embedding methods, such as those proposed by Hamilton[27] allow us to track not the disappearance of a word, but its shift in meaning. How the semantic neighborhood of the word "artificial" changes from 2004 to 2024 in the Wikipedia article "Artificial_intelligence." This would be a qualitative improvement over frequency

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[25] **Wikipedia Editor Survey 2011** was a global survey conducted by the Wikimedia Foundation to examine the demographic characteristics, motivations, editing practices, and experiences of Wikipedia contributors. The survey provided one of the first comprehensive profiles of the editing community, showing that editors were predominantly male, highly educated, and concentrated in North America and Europe.
https://meta.wikimedia.org/wiki/Research%3AWikipedia_Editors_Survey_2011?utm_source=chatgpt.com

[26] **Sinclair, J.** (1991). Corpus, Concordance, Collocation. Oxford UP. Methodology,Corpus Linguistic

[27] **Hamilton, William L., Jure Leskovec, and Dan Jurafsky.** Diachronic Word Embeddings Reveal Statistical Laws of Semantic Change. In Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 1489–1501. Berlin: Association for Computational Linguistics, 2016.
https://doi.org/10.18653/v1/P16-1141

analysis. However, such an analysis requires a significantly larger corpus and computational resources than were available in this study and remains a new direction for future research.
The suppression analysis produced its most interpretively challenging result in the WikiLeaks case. The suppression index of s = 1.97 shows that discourse about transparency, surveillance, and state secrecy nearly doubled. What the index cannot show is whether this increase reflects editorial endorsement of WikiLeaks' mission, neutral documentation of a newsworthy event, or the accumulated output of competing editors with opposing viewpoints. All three scenarios produce the same frequency trajectory. The n-gram method cannot distinguish between them.

The Scale Problem: What Gets Lost in Scale

Michel (2011) worked with five million books. This is both their strength and their weakness. At this scale, patterns become visible that are impossible to detect in smaller corpora. The gradual regularization of irregular verbs over two hundred years, the acceleration of collective forgetting from one generation to the next. But this same scale obscures what happens within a single text, a single community, or a single historical moment.
P. Juola [28] demonstrates an alternative. Stylometry works at the level of a few thousand words and allows us to answer questions that culturomics fundamentally cannot ask such as, who wrote this specific text? Did the author's style change after a certain event? Do two documents belong to the same source? This is not a competing approach. It is a different scale of observation.
The collective memory analysis showed that Wikipedia does not reproduce the accelerating forgetting curve Michel et al. described. Mention frequencies for most historical target years remained stable across the full 21-year observation window, with no measurable half-life. This intermediate-scale finding is inaccessible both to macro-level corpus analysis like culturomics and to micro-level stylometric work. It opens a specific research question: does Wikipedia preserve historical references through genuine institutional memory, or simply through the absence of editorial motivation to delete them? That question requires methods beyond frequency counting to answer.

Limitations

This section describes three levels of limitations: limitations of Michel (2011) original method, limitations of our experimental corpus, and limitations of the analysis tools.

Limitations of the Original Approach

Culturomics in its original shape operates at the level of form, not meaning. This is not an accidental implementation flaw. It is a structural consequence of the chosen unit of analysis. An n-gram records the occurrence of a sequence of symbols in a text. It says nothing about the context in which this sequence appeared, for what purpose, or how it was perceived by the reader.

[28] **Juola, Patrick**. 2013. "Stylometry and Immigration: A Case Study." In Proceedings of the ACL 2013 Workshop on Computational Linguistics for Literature, 22–30. Association for Computational Linguistics. https://www.researchgate.net/publication/339054633_Stylometry_and_Immigration_A_Case_Study

Michel (2011) did not claim to address this problem. They explicitly limited their study to the question of how often a given n-gram is used. The problem arises when the results are interpreted as evidence of cultural phenomena, rather than simply frequency patterns. The increase in the word "evolution" in Google Books tells us that this word was appearing more frequently in books. Does this indicate growing public interest in evolutionary theory? Not necessarily. It may reflect the growing number of biology textbooks in the corpus. Pechenick[29] showed that this type of conflation of corpus artifact and cultural signal is the main methodological vulnerability of culturomics.

The second limitation concerns reproducibility. The full Google Books Ngram Corpus is not available for download in its raw form. Researchers work with aggregated frequency data via Ngram Viewer or a limited public dataset. This means that independent verification of Michel's(2011) results on the same corpus is fundamentally difficult. In science, reproducibility is the one of the primary criteria for the reliability of results.

Limitations of Our Corpus

This study uses a pilot corpus of Wikipedia articles for the period 2001–2024. This is significantly smaller than the five million books in Michel(2011). The small size of the corpus has specific methodological implications.

The first is the statistical instability of rare words. In a small corpus, a word that appears two or three times a year may exhibit extreme frequency fluctuations simply due to chance. When working with Google Books, the scale smooths out these fluctuations. In our corpus, they remain and can simulate trends that do not exist.

The second is the issue of article selection. Our list of articles was not randomly selected. It is intentionally biased toward topics related to digital culture, technology, and pop culture. This is not a mistake. It is a deliberate decision that allows us to test culturomic methods on the type of material characteristic of the digital age and underrepresented in Google Books. However, this means that our results cannot be generalized to all of Wikipedia or to the entire English language.

The third consequence concerns editorial bias on Wikipedia. As noted in Section 3, Wikipedia's editorial base is historically disproportionately demographically diverse. The collective memory we find in our corpus is the memory of a specific community. It cannot be directly compared to the collective memory of society as a whole, which Michel (2011) attempted to measure using a book corpus.

The fourth limitation is a technical gap in the data. Some articles do not have early revisions available through the MediaWiki API, particularly for the years 2001–2003, when Wikipedia was just beginning to fill up. This creates an asymmetry, later years are better covered than earlier ones, which may distort the dynamics of lexicon growth at the beginning of the time series.

Limitations of Analytical Tools

The lexicon "dark matter" check in this study was conducted using WordNet, not the Oxford English Dictionary used by Michel (2011). It is not an equivalent replacement. WordNet is a

[29] **Pechenick, E. A.,** Danforth, C. M., & Dodds, P. S. (2015). "Characterizing the Google Books Corpus: Strong Limits to Inferences of Socio-Cultural and Linguistic Evolution." PLOS ONE.

semantic network organized around concepts, containing approximately 155,000 lemmas. The OED is a historical dictionary with over 600,000 entries and etymological data. WordNet does not index many words found in the OED, and conversely, it includes technical NLP terms not found in the OED. This means that our estimate of the percentage of "dark matter" is systematically overestimated compared to the results of Michel(2011). Additional validation via the Free Dictionary API partially compensates for this, but does not eliminate the fundamental inconsistency between the tools.

A second tool limitation concerns wikitext cleanup. Our clean_wikitext function removed Wikipedia markup to produce clean text. This process is not perfect. Some templates, tables, and special constructs may have been partially retained in the text. This introduces noise into the frequency data, especially for early years when Wikipedia markup was less standardized.

A third limitation concerns the TTR metric. Type-Token Ratio depends on text length where longer texts consistently show lower TTR. Wikipedia articles grow over time. This means that the decline in TTR in later years may simply reflect an increase in article length, rather than a decrease in lexical diversity.

Conclusion

This essay began with a question that is easier to state than to answer. Do culturomics methods work on a corpus that was not designed for them? The answer, based on three methods applied to 85 Wikipedia articles over 21 years, is that they work differently in each case. This is informative rather than disappointing.

Lexical growth holds. The Wikipedia corpus confirms Michel's (2011) core claim that language expands continuously and a significant portion of that expansion remains undocumented in standard reference tools. The 34% of words absent from WordNet, while not directly comparable to the 52% Michel (2011) found against the OED, points to the same phenomenon. New vocabulary emerges faster than lexicographic resources can absorb it. The lower Heaps' Law parameter captures something real about how encyclopaedic writing differs from general prose.

Collective memory transforms. Wikipedia does not forget in the way printed books do. This is not a failure of the method but a discovery about the medium. A living document preserves historical references through editorial inertia rather than active recollection. The absence of a decaying frequency curve is as meaningful as its presence. It suggests that the model of collective forgetting Michel (2011) proposed may be specific to fixed-text corpora, and that digital encyclopaedic corpora requires a different theoretical framework.

The suppression index partially transfers. WikiLeaks and Climate Change both produced measurable signals. Neither approached Michel's(2011) strong suppression threshold. This too is a finding that Wikipedia's distributed editorial consensus produces gentler, slower, and more ambiguous discourse shifts than state-level censorship does. The same mathematical formula captures something real in both cases, but what it captures is not the same thing.

The broader lesson is not that culturomics has limits. Every method has limits. The lesson is that making those limits explicit is itself a form of knowledge. Michel (2011) built a signal detector.

This essay has tested that detector on a different kind of signal and found that the reading requires recalibration. Scale is not insight by itself. The contribution of digital humanities is not to produce more data but to ask better questions about what any given dataset can and cannot tell us.

The three critical arguments developed in Section 3 follow a deliberate progression that is worth making explicit here. Pechenick (2015) attacked the corpus with the argument that Google Books is a biased archive that mistakes the structure of university libraries for the shape of culture. Wikipedia solves that specific problem by offering a genre-homogeneous corpus with transparent editorial provenance, but introduces a different bias in its place. Sinclair (1991) attacked the method with the argument that n-grams count forms without reading meaning, and the WikiLeaks result (s = 1.95) demonstrates this limitation precisely and the index detects that discourse doubled but cannot explain whether editors were endorsing, documenting, or simply expanding encyclopaedic coverage of a contested event. Juola (2013) introduced the scale problem with proclamation that macro-level culturomics makes individual texts invisible, and micro-level stylometry makes corpus-wide patterns invisible. Wikipedia's intermediate scale produced the collective memory finding that neither approach could have generated. Each critique builds on the previous one. Together they do not dismantle culturomics. They locate it within a landscape of methods, each with its own resolution and its own blind spots, and they suggest that the most productive direction for Computational Linguistics is not to choose between scales but to move deliberately between them.

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